PHYSICS

1) If $\lambda = ax$ is linear mass density of a rod of length L on x-axis with one end at x = 0, then x co-ordinate of C.O.M. of rod will be:

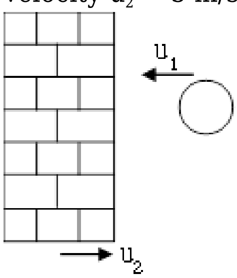
(1) $\frac{3}{2}L$

(2) $\frac{2}{3}L$

(3) $\frac{L}{2}$

(4) None

2) A particle of mass m moves with velocity u_1 = 20 m/s towards a heavy wall that is moving with velocity u_2 = 5 m/s as shown. If collision is elastic then find velocity of particle after collision (in m/s)



(1) 30

(2) 10

(3) 20

(4) 25

3) Body A of mass 4m moving with speed *u* collides with another body B of mass 2m, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is :

(1) $\frac{1}{9}$

(2) $\frac{8}{9}$

(3) $\frac{4}{9}$

(4) $\frac{5}{9}$

4) A metal ball falls from a height of 32 metre on a steel plate. If the coefficient of restitution is 0.5, to what height will the ball rise after second bounce ?

(1) 2 m

(2) 4 m

(3) 8 m

(4) 16 m

5) A ball of mass m moving at a speed V makes a head on collision with an identical ball at rest. The kinetic energy of the balls after the collision is $\left(\frac{3}{4}\right)^{th}$ of the initial kinetic energy then coefficient of restitution for the collision is :

(1) $\frac{1}{2}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{\sqrt{2}}$

(4) $\frac{1}{4}$

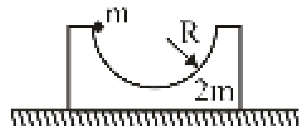
6) A circular plate of uniform thickness of diameter 56 cm, whose center is at origin. A circular part of diameter 42 cm is removed from one edge. What is the distance of the centre of mass of the remaining part

(1) 3 cm

(2) 6 cm

(3) 9 cm

(4) 12 cm

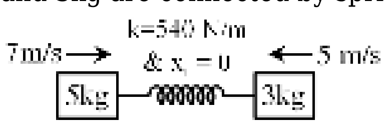
7) A small ball of mass m is released from rest from the position shown. All contact surfaces are smooth. The speed of the ball when it reaches its lowest position is : 

(1) $\sqrt{\frac{2gR}{3}}$

(2) $\sqrt{\frac{4gR}{3}}$

(3) $\sqrt{gR}$

(4) $\sqrt{2gR}$

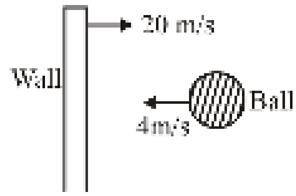
8) Two blocks of mass 5kg and 3kg are connected by spring as shown in figure. Find maximum compression in the spring. 

(1) 0.5

(2) 0.6

(3) 0.7

(4) 0.4

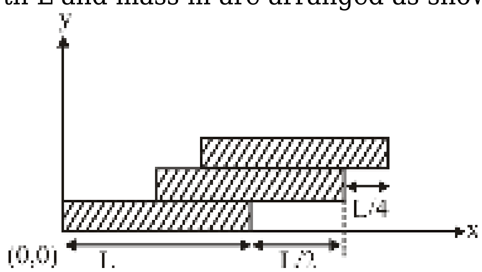
9) Find velocity of ball after collision, if $e = \frac{1}{2}$? 

(1) 32 m/s

(2) 16 m/s

(3) 8 m/s

(4) 4 m/s

10) Three bricks each of length L and mass m are arranged as shown in figure. The distance of the centre of mass of system is :- 

(1) $\frac{L}{4}$

(2) $\frac{L}{2}$

(3) $\frac{3L}{2}$

(4) $\frac{11L}{12}$

11) The kinetic energy K of a particle moving along a circle of radius R depends on the distance covered s as $K = as^2$. The centripetal force acting on the particle is

(1) 2asR

(2) $2as^2$

(3) 2as

(4) $\frac{2as^2}{R}$

12) A particle of mass m starts moving in a circular path of constant radius r, such that its centripetal acceleartion a_c is varying with time t as $a_c = k^2rt^2$, where k is a constant. What is the power delivered to the particle by the force acting on it :

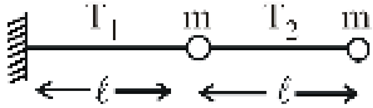
(1) mkr^2t

(2) $m^2k^2r^2t^2$

(3) mk^2r^2t

(4) None of these

13) Two particles having same mass are connected with strings as shown.
It they are rotated in horizontal plane with angnlar

velocity ω, then what will be value of $\frac{T_1}{T_2}$: 

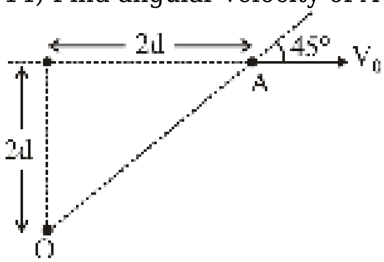
(1) $\frac{1}{4}$

(2) $\frac{1}{2}$

(3) $\frac{3}{2}$

(4) 2

14) Find angular velocity of A with respect to O, at the instant shown in figure.



(1) $\frac{V_0}{d}$

(2) $\frac{V_0}{2d}$

(3) $\frac{V_0}{4d}$

(4) $\frac{V_0}{3d}$

15) A car is negotiating a curved road of radius R. The road is banked at angle θ. The coefficient of friction between the tyres of the car and road is μ. The maximum safe speed on this road is.

(1) $\sqrt{\frac{gR(\mu_s + \tan\theta)}{(1 - \mu_s \tan\theta)}}$

(2) $\sqrt{\frac{g}{R}\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$

(3) $\sqrt{\frac{g}{R^2}\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$

(4) $\sqrt{gR^2\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$

16) A car is moving with speed 20 m/s on a circular path of radius 100 m. Its speed is increasing at

the rate of 3 m/s^2. The magnitude of acceleration of the car at that moment is

(1) 1 m/s^2

(2) 3 m/s^2

(3) 4 m/s^2

(4) 5 m/s^2

17) Keeping the angle of banking same, if the radius of curvature is made four times, the percentage increase in the maximum speed with which a vehicle can travel on a circular road is :–

(1) 25%

(2) 50%

(3) 75%

(4) 100%

18) The displacement x of a particle at a time t is given by x = 5 sin 2t where x is in meter and t is in second. A simple pendulum has the same period as the particle when the length of the pendulum is

(1) 10.0 m

(2) 5.0 m

(3) 2.5 m

(4) 2.0 m

19) A particle is moving along the x-axis under the influence of a force given by F = –5x + 15. At t = 0, the particle is located at x = 6 and is having zero velocity. It takes 0.5 seconds to reach the origin for the first time. The equation of motion of the particle can be represented by

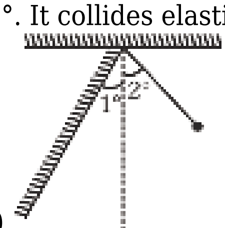
(1) x = 3 + 3 cos π t

(2) x = 3 cos π t

(3) x = 3 + 3 sin π t

(4) x = 3 + 3 cos (2 π t)

20) A simple pendulum of length 1m is allowed to oscillate with amplitude 2°. It collides elastically



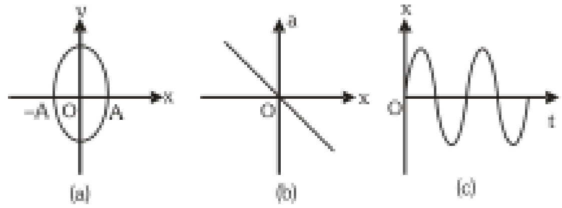
with a wall inclined at 1° to the vertical. Its time period will be : (use g = π^2)

(1) 1/3 sec

(2) 4/3 sec

(3) 2 sec

(4) None of these

21) For a particle excuting S.H.M.  Select the correct graph.

(1) Only a
(2) Only b
(3) Both (a) & (b)
(4) All (a), (b) & (c)

22) The K.E. and P.E of a particle executing SHM with amplitude A will be equal when its displacement is

(1) $\sqrt{2}A$
(2) $\frac{A}{2}$
(3) $\frac{A}{\sqrt{2}}$
(4) $\sqrt{\frac{2}{3}}A$

23) A spring of force constant 'k' is cut into two parts whose lengths are in the ratio 1 : 2. The two parts are now connected in parallel and a block of mass 'm' is suspended at the end of the combined spring. Find the period of oscillation performed by the block.

(1) $T = \pi\sqrt{\left(\frac{2m}{9k}\right)}$
(2) $T = 2\pi\sqrt{\left(\frac{2m}{9k}\right)}$
(3) $T = 2\pi\sqrt{\left(\frac{m}{9k}\right)}$
(4) $T = 2\pi\sqrt{\left(\frac{2m}{k}\right)}$

24) Sound waves from a whistle of frequency 1200 Hz reach a point by two different path. When the paths differ by 10 cm or 40 cm, there is silence at that point. The speed of sound is :-

(1) 100 × 15 cm/sec
(2) 1200 × 30 cm/sec
(3) 1100 × 45 cm/sec
(4) 1200 × 60 cm/sec

25) An organ pipe P_1 closed at one end and vibrating in its first overtone and another pipe P_2 open at

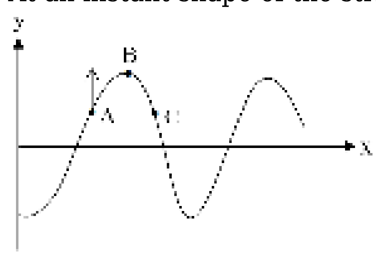
both end vibrating in its second overtone are in resonance. The ratio of length of pipe P_1 and P_2 is :-

(1) 1
(2) 1/2
(3) 1/3
(4) 3/4

26) Equation of travelling wave is given by
$y = 10\sin\left[2\pi t - \frac{\pi}{2}x + \frac{\pi}{6}\right]$m.
the ratio of maximum particle velocity to the wave velocity.

(1) 5
(2) 20π
(3) $\frac{1}{5\pi}$
(4) 5π

27) A wave is travelling along a string. At an instant shape of the string is as shown in figure. At this



instant point A is moving upward then a) The wave is travelling to the right.
b) At this instant C is also moving upward.
c) Amplitude of the wave is equal to the displacement of point B from mean position at this instant.
d) The wave is travelling to the left
Select the correct statements

(1) a, b, c, d
(2) c, d
(3) b, d
(4) a, b, c

28) The first two lengths of an air column, in resonance tube experiment were found to be 32.1 cm and 99.2 cm Determine the end correction for the tube and frequency of the vibrating tuning fork. (V_{Sound} = 332 m/s)

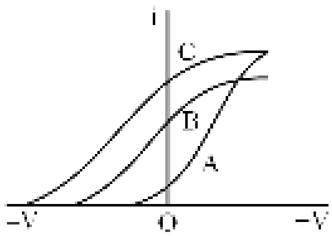
(1) 1.45 cm, 260 Hz
(2) 1.45 cm, 270 Hz
(3) 247.3 Hz, 1.45 cm
(4) 247.3 Hz, 2.45 cm

29) A steel wire of mass 4 gm and length 80 cm is fixed at the two ends. The tension in the wire is 50 N. Find the frequency of the third harmonic.

(1) 62.5 Hz
(2) 187.5 Hz
(3) 125 Hz
(4) 250 Hz

30)

The following graph is between anode potential and photoelectric current in photoelectric experiment for three different light radiations then -



(1) Wave length of A is less than that of B
(2) Wave length of C is maximum while that of A is minimum
(3) Wave length of B is more than C but less than A
(4) Intensity of A is less than C

31) A source of power 15 mW is emitting light of wave length 440 nm. If source capable to convert 10% electrical energy into light then number of quanta emitted by the source per minute.

(1) 1.98×10^{17}
(2) 3.2×10^{15}
(3) 5.4×10^{10}
(4) 1.6×10^{12}

32) Light of wave length 500 nm incident on photo cathode in photoelectric effect experiment work function is 1.2 eV. If anode is kept at 4 volt find kinetic energy of fastest and slowest electrons reaching anode -

(1) 6.28 eV and 3 eV
(2) 5.28 eV and 4 eV
(3) 3.52 eV and 4 eV
(4) 1.42 eV and 0.5 eV

33) In hydrogen atom transition of electron take places from second excited state to ground state, photon so produced incident on photo-cathode and stopping potential found to be 8.09 volt. The threshold frequency of the photocathode is -

(1) 4×10^{15} Hz
(2) 10^{16} Hz
(3) 1.2×10^{10} Hz
(4) 10^{15} Hz

34) Select correct option :

(1) Photoelectric current is directly proportional to the frequency of light.

(2) Stopping potential is related to average frequency in case of poly chromatic light is used.

(3) Stopping potential is discriminator between wave nature and quantum nature of light.

(4) Kinetic energy of fastest electron is directly proportional to the frequency of light.

35) When wavelength of incident light on a metal is λ than kinetic energy of emitted fastest electron is K_1. If wavelength of light becomes $\frac{\lambda}{3}$ then kinetic energy of emitted fastest electron becomes K_2 then correct relation between K_1 and K_2 is -

(1) $K_2 = 3K_1$

(2) $K_2 < 3K_1$

(3) $K_2 > 3K_1$

(4) $K_2 = 4K_1$

36) Select correct statement:
(a) de-Broglie wave length is associated with the matter particles which are in motion.
(b) de-Broglie wave length is independent of charge of particles.
(c) de-Broglie wave length can be associated with photon.
(d) Larger the linear momentum of a matter particle larger is the de-Broglie wave length.

(1) a, b

(2) a, b, c

(3) a, b, c, d

(4) c, d

37) **Assertion (A) :** If a charged particle is projected in transverse electric field its de-Broglie wavelength remains constant.
Reason (R) : In transverse electric field changed particle have uniform circular motion.

(1) Both assertion and Reason are true and reason is correct explanation of assertion.

(2) Both assertion and Reason are true but reason is not correct explanation of assertion.

(3) Both Assertion and Reason incorrect.

(4) Assertion incorrect while Reason correct.

38) Two balls of mass 2 kg and 4 kg and dropped from top of a tower simultaneously find ratio of their de-Broglie wave length when they have fallen through 80 m –

(1) 1 : 2

(2) 4 : 1

(3) 2 : 1

(4) 1 : 4

39) If energy of an electron is increased by 69% then find percentage change in its de-Broglie wavelength –

(1) 25% decrease

(2) 23% decrease

(3) 64% decrease

(4) 16% decrease

40) In hydrogen atom transition of electron takes place from energy level n = 4 to energy level n =2 therefore a photon is emitted, find recoil speed of the atom, m = mass of H atom, R = Rydbarg constant, h = plank's constant.

(1) $\frac{3Rh}{16m}$

(2) $\frac{5Rh}{9m}$

(3) $\frac{2Rh}{3m}$

(4) $\frac{15Rh}{16m}$

41) For nuclear force which of the following is/are correct.
(a) Nuclear force is independent of charge of nucleons.
(b) Nuclear force possess saturation properties.
(c) Nuclear force depends on mutual orientation of spin of nucleons.
(d) Range of nuclear force is Twice that of coulomb force.

(1) Only a

(2) a, c, d

(3) a, b, c

(4) b, c, d

42) A nucleus $^{A}_{Z}X$ decays by emitting two alpha particles, $2\beta^-$ and $4\beta^+$ then ratio of number of neutrons to number of protons in the final nucleus.

(1) $\frac{A-Z-9}{Z-3}$

(2) $\frac{A-Z-1}{Z-8}$

(3) $\frac{A-Z-2}{Z-6}$

(4) $\frac{A-Z+1}{Z-6}$

43) A nucleus at rest splits into two parts having speeds in ratio 1 : 27, then ratio of their nuclear radii will be -

(1) 2 : 1

(2) 1 : 2

(3) 3 : 1

(4) 1 : 3

44) A nucleus of $^{210}_{84}Po$ originally at rest emits α-particle with speed v. What will be the recoil speed of the daughter nucleus –

(1) $\frac{4v}{206}$

(2) $\frac{4v}{214}$

(3) $\frac{v}{206}$

(4) $\frac{v}{214}$

45) The nuclear radius of $_8O^{16}$ is 3×10^{-15} m, if an atomic mass unit is 1.67×10^{-27} kg then the nuclear density is -

(1) 2.35×10^{17} gm/cm^3

(2) 2.35×10^{17} kg/m^3

(3) 2.35×10^{17} gm/m^3

(4) 2.35×10^{17} kg/cm^3

CHEMISTRY

1) How many of the following is correct ?
(A) Rhombic sulphur is insoluble in water but dissolves to some extent in benzene.
(B) Below 369 K , β-sulphur transforms into α-sulphur.
(C) Both Rhombic and monoclinic sulphur have S_8 molecules
(D) α-sulphur is insoluble in CS_2

(1) 4

(2) 2

(3) 1

(4) 3

2) Select the incorrect statement about V group hydrides NH_3, PH_3, SbH_3, BiH_3

(1) The thermal stability of hydrides decreases from NH_3 to BiH_3

(2) The reducing character of hydrides increases from NH_3 to BiH_3

(3) Bond angle increases from NH_3 to BiH_3

(4) Lewis basic character decreases from NH_3 to BiH_3

3) Correct formula of apatite family is :
(where X = F, Cl , OH)

(1) $Ca_3(PO_4)_2.CaX_2$

(2) $2Ca_3(PO_4)_2.CaX_2$

(3) $Ca_3(PO_4)_2.2CaX_2$

(4) $3Ca_3(PO_4)_2.CaX_2$

4) **Assertion (A) :** Oxygen is more electronegative than sulphur, yet H_2S is acidic, while H_2O is neutral.
Reason (R) : H–S bond is weaker than O–H bond.

(1) if both (A) and (R) are true and (R) is the correct explanation of (A)
(2) if both (A) and (R) are true but (R) is not correct explanation of (A)
(3) if (A) is true but (R) is false
(4) if (A) is false and (R) is true

5) The INCORRECT statement regarding the structure of C_{60} is :

(1) The six-membered rings are fused to both six and five-membered rings.
(2) Each carbon atom forms three sigma bonds.
(3) The five-membered rings are fused only to six-membered rings.
(4) It contains 12 six-membered rings and 24 five-membered rings.

6) The reaction of white phosphorus with aqueous NaOH gives a gas (G) along with another phosphorus containing compound. The reaction type ; the number of P–H bonds in gas (G) and number of P–H bonds in the other product are respectively

(1) Acid base reaction ; 3 and 1
(2) redox reaction ; 3 and 3
(3) disproportionation reaction ; 3 and 2
(4) disproportionation reaction ; 3 and 3

7) $Br_2 + 2X^-(aq) \rightarrow X_2 + 2Br^-(aq)$ if above reaction is correct, then X is :-

(1) F^-, Cl^-
(2) I^-
(3) F^- Cl^- I^-
(4) Not possible

8) **Assertion :** Noble gases have very low boiling points.
Reason : Noble gases being monoatomic have no interatomic forces except weak dispersion forces.

(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If Assertion is true but reason is false.
(4) If both assertion and reason are false.

9) Which of the following is complex salt ?

(1) $KCl \cdot MgCl_2 \cdot 6H_2O$
(2) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

(3) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

(4) $4KCN \cdot Fe(CN)_2$

10) Which complex is chelating complex ?

(1) Diamminedichloroplatinum (II)

(2) Bis (ethylene diamine) copper (II)

(3) Tetra carbonyl nickel (0)

(4) Hexacyanoferrate (III) ion

11) Match List–I (Complex ions) with List–II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List-I (Complex ions)		**List-II (Number of Unpaired Electrons)**	
A.	$[CrF_6]^{4-}$	i.	One
B.	$[MnF_6]^{4-}$	ii.	Two
C.	$[Cr(CN)_6]^{4-}$	iii.	Three
D.	$[Mn(CN)_6]^{4-}$	iv.	Four
		v.	Five

Code :

(1) A-iv, B-i, C-ii, D-v

(2) A-ii, B-v, C-iii, D-i

(3) A-iv, B-v, C-ii, D-i

(4) A-ii, B-i, C-iii, D-v

12) Given below are two statements:
Statement I : $N(CH_3)_3$ and $P(CH_3)_3$ can act as ligands to form transition metal complexes.
Statement II: As N and P are from same group, the nature of bonding of $N(CH_3)_3$ and $P(CH_3)_3$ is always same with transition metals.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Statement I is incorrect but Statement II is correct

(2) Both Statement I and Statement II are correct

(3) Statement I is correct but Statement II is incorrect

(4) Both Statement I and Statement II are incorrect

13) In which of the following complex EAN rule is followed.

(1) $[Mn(CO)_5]$

(2) $[Ni(DMG)_2]$

(3) $[Mn_2(CO)_{10}]$

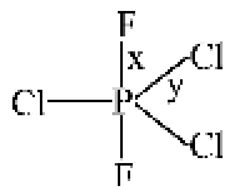
(4) $[V(CO)_6]$

14) Consider $[Co(CO)_4]^-$ as x and $[Mn(CO)_6]^+$ as y to select the incorrect options from the following –

(1) The C – O bond length in x is more than that in y
(2) The C – O bond energy in x is more than that in y
(3) Double bond character in metal to carbon bond in x has more than that in y
(4) Both x and y have the same EAN of the central metal atom

15) The formula of the complex tris(ethylenediamine)cobalt(III)-sulphate is

(1) $[Co(en)_2SO_4]$
(2) $[Co(en)_3SO_4]$
(3) $[Co(en)_3]SO_4$
(4) $[Co(en)_3]_2(SO_4)_3$

16) Correct order of bond length in given compound will be : 

(1) x > y
(2) x < y
(3) x = y
(4) can't say

17) How many compound(s) gives product on hydrolysis in which pπ-dπ bond is present?
XeF_6, N_2O_5, P_4O_6, SF_4, BCl_3

(1) 4
(2) 3
(3) 5
(4) 2

18) If internuclear axis is x-axis then HOMO orbital(s) of O_2 molecule is..

(1) σ^*2p_x
(2) π^*2p_z, π^*2p_y
(3) π^*2p_y, π^*2p_x
(4) $\pi 2p_z$, $\pi 2p_y$

19) In which of the following species all atoms are present in the same plane ?

(1) I_3^-
(2) ICl_4^-

(3) XeF_4

(4) All of the above

20) Find the number of pair(s) having wrong order of indicated property
(i) $SF_6 < TeF_6$ (tendency of hydrolysis)
(ii) $H_2 > He$ (boiling point)
(iii) $BF_3 < BCl_3$ (bond angle)
(iv) $CaCO_3 > BaCO_3$ (thermal stability)
(v) $PbO < PbO_2$ (oxidizing property)

(1) 4

(2) 5

(3) 2

(4) 3

21) Which of the following can show maximum interaction with H_2O ?

(1) Kr

(2) Ar

(3) Xe

(4) All have equal ease

22) Which of the following statements are correct ?

(1) Ionisation energy order; $O_2 > O$

(2) Dimer of NO_2 contains N–O–N linkage

(3) H_3PO_2 is a diprotic acid

(4) All diatomic halogens are coloured due to HOMO-LUMO electron transition.

23) The correct order of 'S—O' bond length is :

(1) $SO_3^{2-} > SO_4^{2-} > SO_3 > SO_2$

(2) $SO_3^{2-} > SO_4^{2-} > SO_2 > SO_3$

(3) $SO_4^{2-} > SO_3^{2-} > SO_2 > SO_3$

(4) $SO_4^{2-} > SO_3^{2-} > SO_3 > SO_2$

24) Match **List-I** with **List-II** :

List-I (Species)		**List-II (Hybrid Orbitals)**	
(a)	SF_4	(i)	sp^3d^2
(b)	IF_5	(ii)	d^2sp^3
(c)	NO_2^+	(iii)	sp^3d

(d)	NH_4^+	(iv)	sp^3
		(v)	sp

Choose the **correct** answer from the options given below :

(1) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)

(2) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)

(3) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)

(4) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

25) In the following the correct bond order sequence is:

(1) $O_2^{2-} > O_2^+ > O_2^- > O_2$

(2) $O_2^+ > O_2^- > O_2^{2-} > O_2$

(3) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(4) $O_2 > O_2^- > O_2^{2-} > O_2^+$

26)

PF_xCl_y is a non polar molecule for which value of x and y :

(1) x = 2, y = 3

(2) x = 3, y = 2

(3) x = 4, y = 1

(4) x = 1, y = 4

27) In which of the following transformations, the bond order has increased and the magnetic behaviour has changed ?

(1) $C_2^+ \rightarrow C_2$

(2) $NO^+ \rightarrow NO$

(3) $O_2 \rightarrow O_2^+$

(4) $N_2 \rightarrow N_2^+$

28) One mole of the complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ion on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of AgCl(s). The structural formula of the complex is -

(1) $[Co(NH_3)_4Cl]Cl_2.NH_3$

(2) $[Co(NH_3)_5Cl].Cl_2$

(3) $[Co(NH_3)_3Cl].2NH_3$

(4) $[Co(NH_3)_4Cl_2]Cl.NH_3$

29) Two electrolytic cells, one containing acidified ferrous chloride and another acidified ferric chloride are connected in series. The ratio of iron deposited at cathodes in the two cells when same electricity is passed through the cells will be :

(1) 3 : 1

(2) 2 : 1

(3) 1 : 1

(4) 3 : 2

30) Equal quantities of electricity are passed through three voltameters containing $FeSO_4$, $Fe_2(SO_4)_3$ and $Fe(NO_3)_3$. Consider the following statements in this regard :-
(a)The amount of iron deposited in $FeSO_4$ and $Fe_2(SO_4)_3$ is equal.
(b) The amount of iron deposited in $Fe(NO_3)_3$ is two third of the amount of Fe deposited in $FeSO_4$
(c) The amount of iron deposited in $Fe_2(SO_4)_3$ and $Fe(NO_3)_3$ is equal
Of these statement :

(1) (a) alone is correct

(2) (a) and (b) are correct

(3) (b) and (c) are correct

(4) (c) alone is correct

31) Calculate cell potential at 298 K for following galvanic cell
$Cd(s)\,|Cd^{+2}(aq.)\,(0.1M)\,||H^+_{(aq)}\,(0.1M)\,|H_2(g, 0.5atm)|Pt$
$E^o_{Cd^{+2}/Cd} = -0.40$

(1) 0.38

(2) – 0.38

(3) 0.36

(4) – 0.36

32) Calculate $E°_{cell}$ for following cell
$Zn(s)\,|Zn^{+2}_{(aq.)}||Ag^+_{(aq)}|Ag(s)$
$E^o_{Zn^{+2}/Zn} = -0.76V$; $E^o_{Ag^+/Ag} = 0.80V$

(1) 0.04 V

(2) –1.56 V

(3) 1.56 V

(4) 0.84 V

33) When aq. solution of $Cr_2(SO_4)_3$ is electrolysed by Pt electrodes, the products formed at the cathode and anode, respectively are :-

(1) Cr and O_2

(2) H_2 and O_2

(3) Cr and SO_2

(4) H_2 and SO_2

34) The standard electrode potential $E^o_{I_2/I^-}$, $E^o_{Br^-/Br_2}$ and $E^o_{Fe/Fe^{2+}}$ are respectively +0.54 V, –1.09 V and 0.44 V. On the basis of above data which of the following process is non-spontaneous

(1) $Br_2 + 2I^- \rightarrow 2Br^- + I_2$

(2) $Fe + Br_2 \rightarrow Fe^{2+} + 2Br^-$

(3) $Fe + I_2 \rightarrow Fe^{2+} + 2I^-$

(4) $I_2 + 2Br^- \rightarrow 2I^- + Br_2$

35) Standard reduction potential for Fe^{+2} / Fe and Sn^{2+}/Sn electrode are –0.44 V and –0.14 V respectively. For the cell reaction
$Fe^{+2} + Sn \rightarrow Fe + Sn^{+2}$, the standard emf is :

(1) +0.30 V

(2) –0.58 V

(3) +0.58 V

(4) –0.30 V

36) Given that E^0 (Zn^{2+} | Zn) = –0.763 V and E^0 (Cd^{2+} | Cd) = –0.403 V. The EMF of the following cell :-
Zn|Zn^{2+} (0.004 (M)) || (0.2 (M)) Cd^{2+} | Cd

(1) E = –0.36 + [0.059 /2] [log 0.004/0.2)]

(2) E = +0.36 – [0.059 /2] [log 0.004/0.2)]

(3) E = –0.36 + [0.059 /2] [log 0.2/0.004)]

(4) E = +0.36 – [0.059 /2] [log 0.2/0.004)]

37) In the electrochemical reaction $2Fe^{+3} + Zn \rightarrow Zn^{+2} + 2Fe^{+2}$
increasing the concentration of Fe^{+2} :-

(1) Increase the cell EMF

(2) Increase the current flow

(3) Decrease the cell EMF

(4) Alter the pH of the solution

38) The cell reaction
$2Ag^+$ (aq) + H_2(g) → $2H^+$(aq) + 2Ag(s), is best represented by :-

(1) Ag(s) | Ag^+ (aq) | | H^+ (aq) | H_2(g) | Pt (s)

(2) Pt(s) | H_2(g) | H^+ (aq) | | Ag^+ (aq) | Ag(s)

(3) Ag(s) | Ag^+ (aq) | | H_2(g) | H^+ (aq) | Pt(s)

(4) Ag^+ (aq) | Ag(s) | | H_2(g) | H^+ (aq)

39) What will be the value of equilibrium constant of the reaction :-
Cu(s) + $2Ag^+$(aq) → Cu^{++}(aq) + 2Ag(s)
E^0_{cell} = 0.48 V
$\left\{\frac{2.303RT}{F} = 0.06\right\}$

(1) 10

(2) 10^{48}

(3) 10^{16}

(4) 10^{24}

40) Which of the following statement is incorrect regarding lead-storage battery.

(1) During discharging, lead act as an anode.

(2) During charging, PbO_2 is obtained at anode.

(3) Density of H_2SO_4 solution increases during discharge.

(4) It is used in automobiles & inverter.

41) **Assertion :** Conductivity of all electrolytes decreases on dilution.
Reason : On dilution number of ions per unit volume decreases.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

42) Consider the following relations for emf of a electrochemical cell :
(a) emf of cell =(Oxidation potential of anode) – (Reduction potential of cathode)
(b) emf of cell = (Oxidation potential of anode) + (Reduction potential of cathode)
(c) emf of cell = (Reduction potential of anode) + (Reduction potential of cathode)
(d) emf of cell = (Oxidation potential of anode) – (Oxidation potential of cathode)
Which of the above relations are correct :

(1) (a) and (b)

(2) (c) and (d)

(3) (b) and (d)

(4) (c) and (a)

43) Limiting molar conductivity of NH_4OH $\left(\text{i.e.} \overset{\circ}{\Lambda}_{m(NH_4OH)}\right)$ is equal to:-

(1) $\overset{\circ}{\Lambda}_{m(NH_4OH)} + \overset{\circ}{\Lambda}_{m(NH_4Cl)} - \overset{\circ}{\Lambda}_{m(HCl)}$

(2) $\overset{\circ}{\Lambda}_{m(NH_4Cl)} + \overset{\circ}{\Lambda}_{m(NaOH)} - \overset{\circ}{\Lambda}_{m(NaCl)}$

(3) $\overset{\circ}{\Lambda}_{m(NH_4Cl)} + \overset{\circ}{\Lambda}_{m(NaCl)} - \overset{\circ}{\Lambda}_{m(NaOH)}$

(4) $\overset{\circ}{\Lambda}_{m(NaOH)} + \overset{\circ}{\Lambda}_{m(NaCl)} - \overset{\circ}{\Lambda}_{m(NH_4Cl)}$

44) In the electrochemical cell :-
$Zn|ZnSO_4(0.01M)||CuSO_4(1.0\ M)|Cu$, the emf of this Daniel cell is E_1. When the concentration of $ZnSO_4$ is changed to 1.0M and that of $CuSO_4$ changed to 0.01M, the emf changes to E_2.

Which one of the relationship is correct between E_1 and E_2?

(1) $E_1 < E_2$
(2) $E_1 > E_2$
(3) $E_2 = 0 \neq E_1$
(4) $E_1 = E_2$

45) Which of the following is incorrect regarding salt bridge solution ?

(1) Solution must be of strong electrolyte
(2) Solution should be inert towards both electrodes
(3) Ionic mobility of cation and anions of salt should be much different
(4) Salt bridge solution is prepared in gelatin or agar-agar to make it semi solid.

BIOLOGY

1) Select the incorrect statement.

(1) Ammonia produced by metabolism is converted to urea in the kidney
(2) Birds excrete nitrogenous waste in the form of uric acid
(3) Aquatic insects are ammonotelic in nature
(4) Kidneys do not play any significant role in the excretion of ammonia in many bony fishes

2) Which of the following statement is incorrect ?

(1) Mammals have the ability to produce a concentrated urine.
(2) The henle's loop and peritubulay capillary play impontent role in production of concentrated urine.
(3) The counter current mechanism Helps to maintain a concentration gradient in the medullary interstitium.
(4) Humans kidneys can produce urine hearty four times concentrated than the initial filtrate formed

3) Which of the following statements are incorrect ?
(a) The first step in urine formation is ultrafiltration of blood.
(b) The endothelium of bowman's capsule leave some minute spaces called filtration slits.
(c) The ascending limb is allows transport of electrolytes by actively only.
(d) mammals have the ability to produce a concentrated urine.

(1) Only b
(2) b, c and d
(3) a, b, and c
(4) b and c

4) **Statement I** :- Nearly all of the essential nutrients, and 70-80 per cent of electrolytes and water are reabsorbed by PCT.

Statement II :- ANF can cause vasodilation and thereby decrease the blood pressure for increase GFR.

(1) Both statement are correct.
(2) Only statement I is correct.
(3) Only statement II is correct.
(4) Both statement are incorrect.

5)

Match the column-I with column-II and choose correct answer.

	Column-I		**Column-II**
A	DCT	i	Filtration of blood
B	Counter current mechanism	ii	Special sensitive region
C	JGA	iii	Maintain concentration gradient of medullary interstitium
D	Malpighian body	iv	Site of conditional reabsorption of Na^+ and water

(1) A-ii, B-i, C-iii, D-iv
(2) A-iv, B-ii, C-iii, D-i
(3) A-iv, B-iii, C-ii, D-i
(4) A-ii, B-iii, C-i, D-iv

6) Mostly aquatic organism are ammonotelic in nature because?

(1) Ammonia is most toxic form and requires large amount of water for it's elimination
(2) Ammonia is insoluble in water
(3) Ammonia can be removed with a minimum loss of water
(4) aquatic animals only produce ammonia during metabolic activity

7) Select the correct statement regarding kidney :-

(1) 10 - 12 cm in width
(2) 2 - 3 cm in width
(3) 220gm - 370 gm in weight
(4) 5 - 7 cm in width

8) Which of the following statement is incorrect ?

(1) Inflammation of glomeruli of kidney is glomercuoneptiritis
(2) Insoluble mass of crystalised salts formed with in the kidney is called uremia.
(3) Kidney transplantation is the ultimate method in the correction of acute renal failure.
(4) Modern clinical procedures have increased the success rate of kidney transplantation.

9) Read the given statements :
(A) On inner convex surface of kidney a notch is present called hilum

(B) The cortex extends in between the medullary pyramids as renal column called columns of Bertini
(C) A fall in glomerular blood pressure can activate the JG cells to release rennin.
(D) Sweat produced by the sweat glands is a watery fluid containing NaCl, small amounts of urea, lactic acids etc.
Select the correct answer :

(1) B and C are correct
(2) A and C are incorrect
(3) C and D are correct
(4) A and D are incorrect

10) Which one of the following pair is incorrect :

(1) Uricotelic .... Birds
(2) Ureotelic .... Insects
(3) Ammonotelic .... Tadpole
(4) Ureotelic .... Elephant

11) Which statement is not correct about glomerular filteration :

(1) On an average 1100 – 1200 ml of blood is filtered by both kidney per minute
(2) Glomerular filtrate is blood plasma except protein
(3) GFR in a healthy individual is approximately 125 ml/ minute or 180 lit./day
(4) Complete blood plasma is filtered in ultrafiltration process

12) Which of the following causes an increase in sodium reabsorption in distal convoluted tubule?

(1) Increase in aldosterone levels
(2) Increase in STH hormone levels
(3) Decrease in aldosterone levels
(4) Decrease in antidiuretic hormone levels

13) Which of the following helps to maintain a concentration gradiant in the medullary interstitium.

(1) JGA
(2) ANF mechanism
(3) Counter current mechanism
(4) Renin Angiotensin mechanism

14) **Assertion (A) :-** There is no backflow of urine from urinary bladder to kidney.
Reason (R) :- Oblique opening of ureter into the bladder is present.

(1) Both A & R are true & R is correct explanation of A
(2) Both A & R are true but R is not the correct explanation of A
(3) Assertion is true but reason is not true
(4) Reason is true but Assertion is not true.

15) What is the function of the ureters in the urinary system ?

(1) Filtration of blood
(2) Storage of urine
(3) Transport of urine from kidney to the bladder
(4) Regulation of electrolyte balance

16) **Statement-I** :- Urine formed by the nephrons is ultimately carried to the urinary bladder.
Statement-II :- Other than the kidneys, lungs,liver and skin also help in the elimination of excretory wastes.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

17) A condition called uremia in which malfunctioning of kidneys can lead to accumulation of :-

(1) Stone in blood
(2) Urea in blood
(3) Oxalates in blood
(4) Insoluble mass in blood

18) The ascending limb of loop of Henle (Nephron) is impermeable to water but allows transport of electrolytes :-

(1) Actively only
(2) Passively only
(3) Actively or passively
(4) Above statement is completely wrong

19) Which factor helps in maintaining an increasing osmolarity towards the inner medullary interstitium ?
a. Counter current pattern in vasa recta
b. Counter current pattern in Henle's loop
c. Proximity between the Henle's loop and vasa recta

(1) a and b
(2) a, b and c
(3) c only
(4) a and c

20) During micturtion :-

(1) Urethral sphincter contracts, urinary bladder relaxes
(2) Urethral sphincter and urinary bladder contracts
(3) Urethral sphincter relaxes, urinary bladder contracts

(4) Urethral sphincter and urinary bladder relaxes

21) How many of the following are excreted in urine of human ?
NH_4^+, Urea, Creatinine, Uric acid, Na^+, Water

(1) Two
(2) Four
(3) Six
(4) Zero

22) Fill in the blanks
Inside the kidney, there are two zones an outer __a__ and an inner __ b__.

(1) a - Medulla, b - cortex
(2) a - cortex, b - medulla
(3) a - pelvis, b - medulla
(4) a - medulla, b - pelvis

23) Fill in the blanks
In Human kidney inner to Hilum is a broad funnd shaped space called the ___ with projections called calyces.

(1) Renal pelvis
(2) Renal costex
(3) Renal Medullar
(4) Nephron

24) Which of the following is correct.

(1) Kidney's are close to dorsal inner wall of abdominal cavity
(2) Each kidney of adult human measures 10-12 cm in length, 5-7 cm in width, 2-3 cm in thickness
(3) Average weight of kidney in human 120 -170 gm
(4) All of these

25) **Statement -I** :- Kidneys are reddish brown bean shaped structure.
Statement-II :- Kidneys are situated between the levels of last thoracic and Third lumbar vertebra.

(1) Only I is true
(2) Only II is true
(3) Both I and II are true
(4) Neither I, Nor II is true

26) **Assertion :-** PCT & DCT of cortical nephron & juxta medullary nephron present in cortical region. **Reason :-** In some of the nephron loop of henle is very long & runs deep into the medulla. These nephrons are called jaxtamedullary nephron.

(1) Both Assertion & Reason are False.

(2) Assertion is True but the Reason is False.

(3) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

27) **Statement-I :** Glomerulus along with Bowman's capsule is called Malpighian Tubules.
Statement-II : The Malpighian corpuscle , PCT and DCT of the Nephron are situated in the Medulla region of Kidney.
Options :

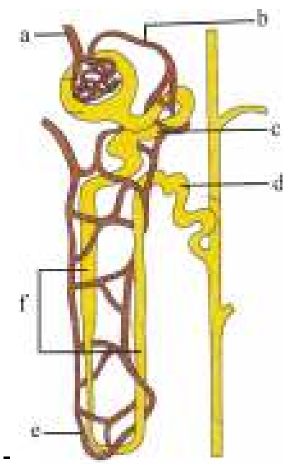
(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

28) Recognise the figure and find out the correct matching :-



(1) c-PCT, d-DCT, a-afferent arteriole, b-efferent arteriole, f–henle's loop, e-vasa recta

(2) d-PCT, c-DCT, b-afferent arteriole, a-efferent arteriole, e-henle's loop, f-vasa recta

(3) c-PCT, d-DCT, b-afferent arteriole, a-efferent arteriole, f–henle's loop, e-vasa recta

(4) d-PCT, c-DCT, a-afferent arteriole, b-efferent arteriole, e-henle's loop, f-vasa recta

29)

Which of the following parts of the nephron given in column A is correctly matched with their functions given in column B ?

	COLUMN (A)		COLUMN (B)
i	Proximal convoluted tubule	a	Sodium is actively reabsorbed in this region
ii	Distal convoluted tubule	b	Sodium and water are reabsorbed under the influence of hormone in this region
iii	Descending limb	c	Primary site of glucose and amino acid reabsorption
iv	Ascending limb	d	Major substances are reabsorbed here is water by osmosis

	i	ii	iii	iv
(1)	a	b	c	d

(2)	b	c	d	a
(3)	c	b	d	a
(4)	d	c	a	b

(1) 1

(2) 2

(3) 3

(4) 4

30)

Which of the following statements are correct?
A. An excessive loss of body fluid from the body switches off osmoreceptors.
B. ADH facilitates water reabsorption to prevent diuresis.
C. ANF causes vasodilation.
D. ADH causes increase in blood pressure.
E. ADH is responsible for decrease in GFR.
Choose the **correct** answer from the options given below:

(1) B, C and D only

(2) A, B and E only

(3) C, D and E only

(4) A and B only

31) In human beings ciliary movement is found in :

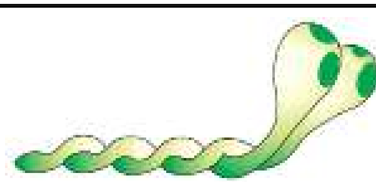
(1) macrophages and Leucocytes

(2) Fallopian tube and Trachea

(3) Small Intestine

(4) Fallopian tube and vasa deferentia

32) Identify the following structure related with muscle shown below and select the correct.
Option for together.

	 (a)	 (b)
(1)	Thin Filament	Actin
(2)	Thick filament	Thin filament
(3)	Thin filament	Meromyosin
(4)	Light Meromyosin	Heavy Meromyosin

(1) 1

(2) 2

(3) 3

(4) 4

33) Which of the following statement are incorrect regarding skeletal muscle ?
(A) Muscle bundles are held together by fascicle
(B) Sarcoplasmic reticulum of muscle fibre is store house of calcium ions.
(C) Striated appearance of skeletal muscle fibre is due to distribution pattern of action & myosin protein
(D) M-line is consider as functional unit of contraction called sarcomere

(1) B & C

(2) A, C & D

(3) A & D

(4) C & D

34) **Statement-I :-** In osteoporosis disorder increased bone mass and increased chances of fractures.
Statement-II :- Myasthenia gravis is a auto immune disorder affecting motor-end plate.

(1) Statement-I is true but statement-II is false

(2) Statement-I is false but statement-II is true

(3) Both statement-I and statement-II are true

(4) Both statement-I and statement-II are false

35) **Statement-I :** During the muscles contraction myosin head binds to the unexposed active site on actin to form a cross bridge.
Statement-II : ATP hydrolysis is required for the breakdown of cross bridge.

(1) Both statement-I and statement-II are correct

(2) Only statement-I is correct

(3) Only statement-II is correct

(4) Both statement-I and statement-II are incorrect

36) Choose the incorect statement from the options given below :-

(1) Vertebrochondral ribs consists of 8^{th}, 9^{th} and 10^{th} pairs

(2) Floating ribs consists of 11^{th} and 12^{th} pairs

(3) Scapula is a cylindrical bone of pectoral girdal

(4) Coxal bone is made up of ilium, ischium, pubis

37) Match List-I with List-II

List-I		**List-II**	
A	Hinge joint	I	Knee joint
B	Pivot joint	II	Between Atlas and axis

C	Gliding joint	III	Between the carpals
D	Ball and socket joint	IV	Between femur and Pelvic girdle

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-III, D-IV

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-II, C-III, D-I

38) Match List-I with List-II :-

	List-I		**List-II**
(A)	Vertebrochondral ribs	(I)	Ilium, Ischium, Pubis
(B)	Scapula	(II)	8^{th}, 9^{th} and 10^{th} pairs
(C)	Vertebrosternal Ribs	(III)	Acromian
(D)	Hip bone	(IV)	$1\text{-}7^{th}$ pairs

Choose the correct answer from the option given below :-

(1) A-III, B-IV, C-II, D-I

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-I, B-III, C-II, D-IV

39)

Select the **correct** statements regarding mechanism of muscle contraction.
A. It is initiated by a signal sent by CNS via sensory neuron.
B. Neurotransmitter generates action potential in the sarcolemma.
C. Increased Ca^{++} level leads to the binding of calcium with troponin on actin filaments.
D. Masking of active site for actin is activated.
E. Utilising the energy from ATP hydrolysis to form cross bridge.
Choose the **most appropriate** answer from the options given below:

(1) B, C and E only

(2) C, D and E only

(3) A and D only

(4) B, D and E only

40) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Calcium plays a important role in skeletal muscle contraction.
Reason (R) : Ca^{+2} binds to subunit of troponin to remove the masking of active sites for myosin on actin filaments.
In the light of the above statements, choose the most appropriate answer form the options given below.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is incorrect

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

41) Examine the figure of pectoral girdle and forelimb and identify the parts labelled as A,B, C and D

	A	**B**	**C**	**D**
(1)	Clavicle	Humerus	Ulna	Carpals
(2)	Scapula	Femur	Radius	Tarsals
(3)	Clavicle	Femur	Radius	Carpals
(4)	Scapula	Humerus	Ulna	Tarsals

(1) 1

(2) 2

(3) 3

(4) 4

42) Which statement is incorrect :

(1) The skull is composed of two sets of bones cranial and facial, that totals to 22 bones

(2) Axial skeleton comprises 80 bones

(3) Axial skeleton comprises the skull, vertebral column, girdle and ribs

(4) The facial region is made up of 14 skeletal elements

43) Choose the correct statement :
a- Dark band is called A-band
b- Cardiac muscles are striated
c- Z line bisects I-band

(1) a and b only

(2) b and c only

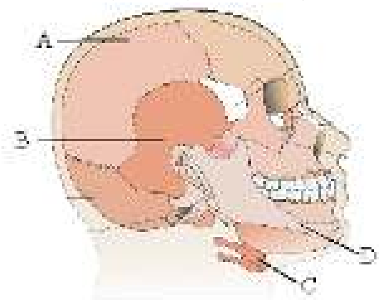
(3) a and c only

(4) a, b and c

44) Mark the correct option for cardiac muscle

(1) These are striated on the basis of structure
(2) These are non-striated on the basis of structure
(3) These are voluntary on the basis of function
(4) Intercalated discs are not present in these muscles

45) Study the given diagrammatic view of human skull and select the correct option.



(1) A–Parietal bone, that is a facial bone
(2) B–Occipital bone, that is a cranial bone
(3) C–Hyoid bone, present at the base of the pharyx
(4) D–Mandible, that is a facial bone

46) The experiment in which a part of leaf is enclosed in a test tube containing KOH soaked cotton, proves that:

(1) Light is essential for photosynthesis
(2) CO_2 is essential for photosynthesis
(3) O_2 releases during photosynthesis
(4) Chlorophyll is essential for photosynthesis

47) Engelmann's experiment with *Cladophora* demonstrated that :-

(1) The full spectrum of sunlight is equally effective for photosynthesis
(2) Only red wavelengths are responsible for photosynthesis
(3) Only blue wavelengths are effective in photosynthesis
(4) Both blue and red wavelengths are effective in photosynthesis

48) In the chloroplast, the site of carbon reaction and photochemical reaction, is-

(1) Grana and stroma respectively
(2) Grana and matrix respectively
(3) Matrix and stroma respectively
(4) Stroma and grana respectively

49) In paper chromatography technique chlorophyll-a and xanthophylls respectively shows :-

(1) bright green and yellow colour

(2) bluish green and purple colour
(3) yellow and blue green colour
(4) purple and yellow colour

50) The light harvesting complex (LHC) is made up of :-

(1) Only one molecule of Chl a
(2) Very few molecules of Chl a only
(3) hundreds of pigment molecules bound to proteins
(4) chl a, chl c, protein and DNA

51) Hatch and Slack pathway is otherwise known as C_4 pathway because:-

(1) The first stable product is oxalocacetic acid/OAA which is a C_4 compound
(2) The primary CO_2 acceptor is OAA, a C_4 compound
(3) All intermediate metabolites are C_4 compound
(4) At a time, 4 CO_2 molecules take part in carboxylation pathway

52) Which of the following is formed during photorespiration?

(1) Sugar
(2) Phoshoglycolate
(3) NADPH
(4) ATP

53)

At low light condition, which of the following plants respond to high CO_2 conditions ?

(1) Only C_3 plants
(2) Only C_4 plants
(3) Neither C_3 nor C_4 plants
(4) Both C_3 and C_4 plants

54) Which statement is **incorrect** ?

(1) C_3 plants respond to higher temperature, show higher photosynthesis rate while C_4 plants have lower temperature optimum.
(2) Tropical plants have higher temperature optimum than the plants adapted to temperate climate.
(3) Light reaction is less temperature sensitive than dark reaction.
(4) The effect of water as a factor is more through its effect on plant rather than directly on photosynthesis.

55) Given below are two statements : One is labelled as **Assertion-A** and the other is labelled as **Reason-R**.
Assertion (A) : It is considered that chlorophyll 'a' is the chief pigment associated with

photosynthesis.
Reason (R) : Wavelengths at which there is maximum absorption by chlorophyll 'a' i.e. in the blue and red regions, also shows higher rate of photosynthesis.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both A and R are true but R is NOT the correct explanation of A.
(2) A is true but R is false.
(3) A is false but R is true.
(4) Both A and R are true and R is the correct explanation of A.

56)

	Characteristics	**C_3 plant**	**C_4 plant**
(A)	Primary CO_2 acceptor	(i)	(ii)
(B)	Number of carbons in the primary CO_2 acceptor	(iii)	3
(C)	RuBisCO in mesophyll cells	Yes	(iv)

Choose the correct combination for (i) to (iv):-
(1) (i)–PEP, (ii)–OAA, (iii)–3, (iv)–Yes
(2) (i)–PGA, (ii)–RuBP, (iii)–5, (iv)–No
(3) (i)–RuBP, (ii)–OAA, (iii)–4, (iv)–Yes
(4) (i)–RuBP, (ii)–PEP, (iii)–5, (iv)–No

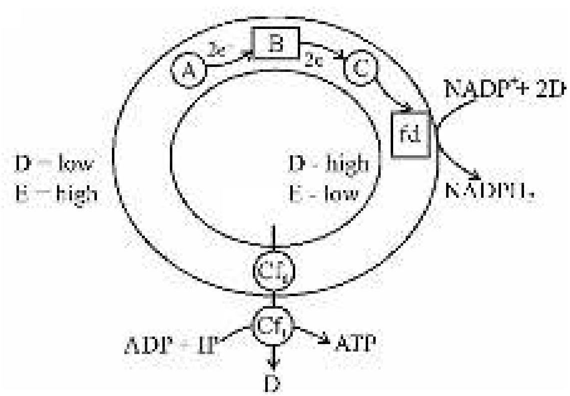
57) **Statement-I :** In green plants H_2O is the hydrogen donor and is oxidised to O_2.
Statement-II : Cornelius Van Niel demonstrated that photosynthesis is essentially a light dependent reaction in which hydrogen from a suitable oxidisable compound oxidise carbon dioxide to carbohydrate.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Statement-I & Statement-II both are incorrect.

58) For 3 CO_2 molecules entering the Calvin cycle, required energy is :-

(1) 3 ATP and 2 NADPH
(2) 2 ATP and 3 NADPH
(3) 9 ATP and 6 NADPH
(4) 2 ATP and 2 NADPH

59) In the given diagram identify A, B, C, D and E correctly when light falls on A and C :-



	A	B	C	D	E
(1)	PSII	NAD^+	P-700	H^+	PH
(2)	PSII	ETS	P-680	H^+	+ve charge
(3)	PSII	ETS	P-700	H^+	PH
(4)	PSII	PQ	P-680	H^+	+ve charge

(1) 1
(2) 2
(3) 3
(4) 4

60) Breaking down of C – C bonds of complex compounds like carbohydrate, lipids & proteins releases considerable amount of energy in the proces of cellular respiration. This released energy:-

(1) Can be used directly for the cellular metabolism.
(2) Cannot be used indirectly for cell activities.
(3) Is used to synthesize ATP.
(4) is completely converted into heat

61) Why plants do not have specialised organs for gaseous exchange ?

(1) Each plant part takes care of its own gas-exchange needs
(2) Plants do not present great demands for gas exchange
(3) The distance that gases must diffuse even in large, bulky plant is not great
(4) All of the above

62) In Glycolysis two redox equivalents are removed from :-

(1) PGAL
(2) BPGA
(3) Pyruvic acid
(4) PEP

63) In eukaryotic respiration, pyruvate dehydrogenase complex is present in :-

(1) Mitochondrial matrix
(2) Inner mitochondrial membrane
(3) Cytoplasm
(4) Perimitochondrial space

64) In Kreb's cycle $FADH_2$ released during conversion of :-

(1) Malic Acid $\longrightarrow$ Oxaloacetic acid
(2) Succinic Acid $\longrightarrow$ malic acid
(3) α-Ketoglutaric $\longrightarrow$ Succinic acid
(4) Citric Acid $\longrightarrow$ α-Ketoglutaric acid

65) Cytochrome C is a small protein attached to -

(1) Inner surface of the Inner membrane of mitochondria
(2) Inner surface of the outer membrane of mitochondria
(3) Outer surface of the outer membrane of mitochondria
(4) Outer surface of the inner membrane of mitochondria

66) Acetyl Co-A is produced by :-

(1) Pyruvic acid
(2) Fatty acid
(3) Protein
(4) All of these

67) When proteins are used as respiratory substrates the value of R.Q. (Respiratory Quotient) is :-

(1) Less than the R.Q. of both fats and carbohydrates
(2) More than the R.Q. of both fats and carbohydrates
(3) Less than the R.Q. of fats but more than the R.Q. of carbohydrates
(4) More than the R.Q. of fats but less than the R.Q. of carbohydrates

68) Conversion of pyruvic acid into ethyl alcohol requires the presence of which enzyme(s) ?

(1) Carboxylase
(2) Dehydrogenase only
(3) Decarboxylase and dehydrogenase
(4) Phosphatase

69) **Assertion :** Under anaerobic conditions, there is net synthesis of only 2 ATP molecules by yeast per glucose molecule.
Reason : In alcohol fermentation, less than seven percent of the energy of glucose is released and not all of it is trapped as ATP.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

70) Select the **incorrect** statement about assumptions of respiratory balance sheet :-

(1) NADH synthesised in glycolysis are transfered into the mitochondria.
(2) None of the intermediates in pathway are utilised to synthesise any other compound.
(3) Only glucose is being respired
(4) Any alternate substrate can enter in pathway at any intermediary stage.

71) Match **List-I** with **List-II** :

List-I		**List-II**	
A.	EMP pathway	I	Release of one CO_2
B.	Tricarboxylic acid cycle	II	Cytochrome C oxidase
C.	Oxidative phosphorylation	III	Release of two CO_2
D.	Oxidative decarboxylation	IV	BPGA → PGA

Choose the correct answer from the options given below :
(1) A-I, B-IV, C-III, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-II, C-IV, D-I
(4) A-IV, B-III, C-II, D-I

72) **Statement-I :-** In TCA cycle, two successive steps of decarboxylation lead to the formation of α-keto glutaric acid and then succinyl-CoA.
Statement-II :- During the conversion of succinyl-CoA to succinic acid a molecule of GTP is synthesised.

(1) Statement I is correct but statement II is incorrect
(2) Both statements are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statements are correct

73) How many cytochromes in the list given below are the member of mitochondrial electron transport system ? Cytochrome d, Cytochrome a, Cytochrome C_1, Cytochrome C_6, Cytochrome a_3, Cytochrome b

(1) Five
(2) Four
(3) Six
(4) Two

74) If a stem cutting is treated with auxin, which of the following is likely to result ?

(1) Root initiation
(2) Suppression of apical dominance
(3) Growth of lateral buds
(4) Bolting of the shoot

75) Growth at cellular level is :-

(1) Mainly increase in protoplasm
(2) Mainly increase in cell number
(3) Mainly increase in cell wall content
(4) Easy to measure directly

76) Maximal size in terms of wall thickening and protoplasmic modifications are achieved by :-

(1) Cells of division phase
(2) Cells of maturation phase
(3) Cells of elongation phase
(4) Cells of meristematic tissue

77) Efficiency index is the measure of :-

(1) Ability of the plant to produce new plant material
(2) Time of growth
(3) Number of cells that lose capacity to divide
(4) Rate of differentiation of plant cells

78) Select out the **correct** match :-

(1) Darwin - Bakanae
(2) Kurosawa - Extract of yeast
(3) Skoog - *Avena* (oat)
(4) Cousins - ABA

79) Which causes fruits like apple to elongate & improve its shape?

(1) Ethylene
(2) Gibberellins
(3) Auxin
(4) ABA

80) Production of cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants ?

(1) Gibberellin
(2) Ethylene

(3) Cytokinin

(4) ABA

81) Which among the following is **not** related to ABA?

(1) Stress hormone

(2) Induces seed dormancy

(3) Induces shoot growth

(4) Act as a general plant growth inhibitor and inhibitor of plant metabolism

82) **Assertion :** Development is a term that includes all changes that an organism go through its life cycle.
Reason : Plants follows different pathways in response to environment or phases of life to form different kinds of structures.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

83) Find the **correct** match :-

<table>
<tr><td rowspan="4">(1)</td><td>Auxin</td><td>Apical dominance</td></tr>
<tr><td>Gibberellins</td><td>Flowering in pineapple</td></tr>
<tr><td>Cytokinins</td><td>Formation of new leaves</td></tr>
<tr><td>ABA</td><td>Promotes Seed dormancy</td></tr>
<tr><td rowspan="4">(2)</td><td>Auxin</td><td>Parthenocarpy</td></tr>
<tr><td>Gibberellins</td><td>Bolting</td></tr>
<tr><td>Cytokinins</td><td>Promotion of nutrient mobilisation</td></tr>
<tr><td>ABA</td><td>Closure of stomata</td></tr>
<tr><td rowspan="4">(3)</td><td>Auxin</td><td>Malting</td></tr>
<tr><td>Gibberellins</td><td>Herbicide</td></tr>
<tr><td>Cytokinins</td><td>Flowering in mango</td></tr>
<tr><td>ABA</td><td>Overcomes apical dominance</td></tr>
</table>

(4)	Auxin	Formation of new leaves
	Gibberellins	Closure of stomata
	Cytokinins	Parthenocarpy
	ABA	Bolting

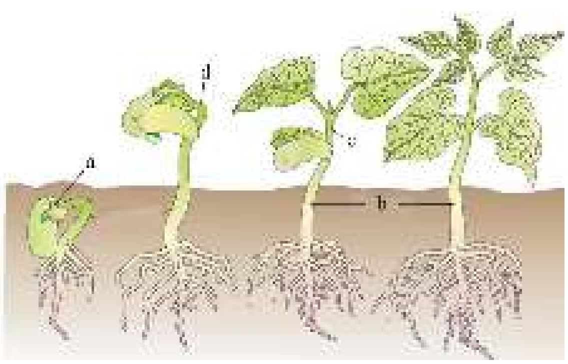
(1) 1
(2) 2
(3) 3
(4) 4

84) **Statement-I :** Growth is regarded as one of the fundamental and conspicuous characteristics of living being.
Statement-II : Growth can be defined as an irreversible permanent increase in size of an organ or its part or even of an individual cell.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

85)

Find out **correct** labelling sequence in given figure:



(1) a = cotyledon, b = hypocotyl, c = epicotyl, d = epicotyl hook
(2) a = Hypocotyl, b = cotyledon, c = epicotyl, d = epicotyl hook
(3) a = Epicotyl, b = epicotyl hook, c = cotyledon, d = hypocotyl
(4) a = cotyledon, b = hypocotyl, c = epicotyl hook, d = epicotyl

86) Choose the **incorrect** statement-

(1) Growth, differentiation and development are very closely related events in the life of a plant

(2) Development in plants is under the control of intrinsic and extrinsic factors both

(3) Heterophylly in cotton, coriander and larkspur are examples of plasticity

(4) Formation of interfascicular cambium and cork cambium from fully differentiated parenchyma cells is termed as redifferentation

87) How many plant growth regulators, in the list given below, are **not** naturally found in plants ?
2,4-Dichlorophenoxy acetic acid, (2,4-D)
Indole butyric acid,
Zeatin,
Naphthalene acetic acid (NAA),
Kinetin,
Indole acetic acid,
Gibberellic acid (GA_3)

(1) Three

(2) Five

(3) Four

(4) Two

88) Some plants show a special type of photosynthesis in which the product of primary CO_2 fixation, that takes place in the mesophyll cells, is not a 3-carbon organic acid and in these plants carbohydrates are synthesised by Calvin pathway but not in the chloroplasts of mesophyll cells. These plants :

(1) respond to increasing CO_2 concentration and saturation is seen only beyond 450 μ□L^{-1}

(2) Can prevent a wasteful oxygenation reaction catalysed by RuBisCO

(3) Show a very lower temperature optimum

(4) can not survive in dry tropical regions

89) **Assertion (A) :** The continued oxidation of respiratory intermediates via the TCA cycle requires the continued regeneration of oxaloacetic acid and also requires regeneration of NAD^+ and FAD^+ from NADH and $FADH_2$ respectively.
Reason (R) : The TCA cycle starts with the condensation of succinyl group with oxaloacetic acid and water. At few points in the cycle NAD^+ and FAD^+ are reduced to NADH and $FADH_2$ respectively.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

90) **Statement-I :** The plant growth regulators are small, simple molecules of diverse chemical composition. Some of these are indole compounds [eg. - Indole-3-acetic acid], derivatives of accessory photosynthetic pigments [eg. :- ABA], terpenes [Gibberellic acid].
Statement-II : The plant growth regulators (PGRs) can be broadly divided into two groups based on their functions in a living plant body. Among these two, PGRs of one group play an important role in plant responses to stresses of biotic and abiotic origin. Most of the PGRs of this group are derivatives of adenine nitrogen base.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**1**	**2**	**1**	**3**	**3**	**2**	**3**	**1**	**4**	**4**	**3**	**3**	**3**	**1**	**4**	**4**	**3**	**4**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**4**	**3**	**2**	**2**	**2**	**4**	**2**	**3**	**2**	**3**	**1**	**2**	**4**	**3**	**3**	**1**	**3**	**3**	**2**	**1**
Q.	41	42	43	44	45															
A.	**3**	**3**	**3**	**1**	**2**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**4**	**3**	**4**	**1**	**4**	**3**	**2**	**1**	**4**	**2**	**3**	**3**	**3**	**2**	**4**	**2**	**2**	**2**	**4**	**4**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**3**	**4**	**2**	**3**	**3**	**1**	**1**	**2**	**4**	**3**	**1**	**3**	**1**	**4**	**4**	**2**	**3**	**2**	**3**	**3**
Q.	86	87	88	89	90															
A.	**1**	**3**	**2**	**2**	**3**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**1**	**2**	**4**	**2**	**3**	**1**	**4**	**2**	**2**	**2**	**4**	**1**	**3**	**1**	**3**	**4**	**2**	**3**	**2**	**3**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**3**	**2**	**1**	**4**	**3**	**3**	**2**	**1**	**3**	**1**	**2**	**3**	**3**	**2**	**4**	**3**	**1**	**3**	**1**	**4**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**1**	**3**	**4**	**1**	**4**	**2**	**4**	**4**	**1**	**3**	**1**	**2**	**3**	**1**	**4**	**4**	**2**	**3**	**3**	**3**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**4**	**1**	**1**	**2**	**4**	**4**	**4**	**3**	**1**	**4**	**4**	**4**	**2**	**1**	**1**	**2**	**1**	**2**	**2**	**2**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**3**	**2**	**2**	**1**	**1**	**4**	**1**	**2**	**2**	**2**										

SOLUTIONS

PHYSICS

1)

Center of mass of a variable-density rod is given by: $x_{cm} = \dfrac{\int_0^L x \cdot \lambda(x)\,dx}{\int_0^L \lambda(x)\,dx}$

where $\lambda(x) = ax$

$$x_{cm} = \frac{\int_0^L x \cdot ax\,dx}{\int_0^L ax\,dx}$$

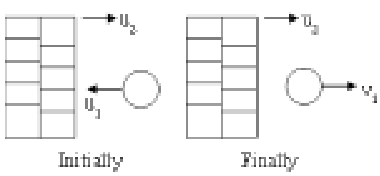
Total Mass = $M = \int_0^L ax\,dx = \dfrac{aL^2}{2}$

$\text{Moment} = \int_0^L ax^2\,dx = \dfrac{aL^3}{3}$

Center of Mass: = $x_{cm} = \dfrac{\frac{aL^3}{3}}{\frac{aL^2}{2}} = \dfrac{2L}{3}$

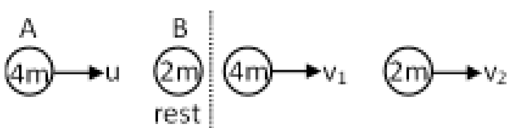
OR

$$x_{cm} = \frac{\int dmx}{\int dm} = \frac{\int (axdx)\,x}{\int axdx} = \left[\frac{\frac{x^3}{3}}{\frac{x^2}{2}}\right]_0^{\ell}$$



2)

$$e = \frac{\text{vel. sep}}{\text{vel. app}} \Rightarrow 1 = \frac{v_1 - 5}{25} \Rightarrow v_1 = 30 \text{ m/s}$$



3)

$$v_1 = \frac{4m - 2m}{4m + 2m}u = \frac{2mu}{6m} = \frac{u}{3}$$

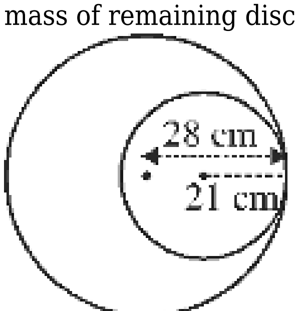
Fraction of energy lost = $\dfrac{\frac{1}{2}(4m)u^2 - \frac{1}{2}(4m)\left(\frac{u}{3}\right)^2}{\frac{1}{2}(4m)u^2} = 1 - \dfrac{1}{9} = \dfrac{8}{9}$

4) $h_n = he^{2n} = 32\left(\frac{1}{2}\right)^4 = \frac{32}{16} = 2$ m
(here n = 2, e = 1/2)

5)

loss in KE = $\frac{1}{2}\frac{m_1 m_2}{(m_1 + m_2)}(\vec{V}_1 - \vec{V}_2)^2$ $(1-e^2)$
$\frac{1}{4}\left(\frac{1}{2}mv^2\right) = \frac{1}{2}\frac{m \times m}{(m+m)}(V^2)(1-e^2)$
$\frac{1}{2} = 1 - e^2 \Rightarrow e = \frac{1}{\sqrt{2}}$

6) Density of disc
$\rho = \frac{m}{\pi \times 28 \times 28 \times t}$
$m_1 = \frac{m}{\pi \times 28 \times 28 \times t} \times \pi \times 21 \times 21 \times t$
$m_1 = \frac{9m}{16}$
mass of remaining disc = m – $\frac{9m}{16} = \frac{7m}{16}$



$\frac{7m}{16}x = \frac{9m}{16} \times 7$
x = 9 cm

7)

Let the speed of ball be v and speed of the wedge be v_2. Using conservation of momentum
$0 = mv + 2mv_2$
$v_2 = -\frac{v}{2}$
Now, using work-energy theorem on the system,
$mgR = \frac{1}{2}mv^2 + \frac{1}{2}(2m) \times \left(\frac{v}{2}\right)^2$
$\Rightarrow v = \sqrt{\frac{4gR}{3}}$

8) At max compression velocity will be same
COLM $\quad 5 \times 7 - 3 \times 5 = (5 + 3)v$
$35 - 15 = 8v$

$$v = \frac{20}{8} = \frac{5}{2} m/s$$

Now COME

$$\frac{1}{2}.5.(7)^2 + \frac{1}{2}.3.(5)^2 = \frac{1}{2}(5+3)\left(\frac{5}{2}\right)^2 + \frac{1}{2}kx^2$$

$$5 \times 49 + 3 \times 25 = 8 \times \frac{25}{4} + kx^2$$

$$320 = 50 + kx^2$$

$$270 = kx^2$$

$$x^2 = \frac{270}{540}$$

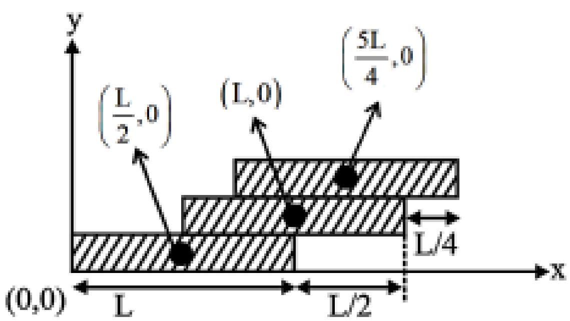
$$x = \frac{1}{\sqrt{2}} = 0.707$$

9) Ball (1) : $u_1 = -4$ m/s, $V_1 = ?$

Wall (2) : $u_2 = +20$ m/s, $V_2 = +20$ m/s

$$e = \frac{1}{2} = \frac{20 - V_1}{-4 - 20}$$

$V_1 = 32$ m/s

10) 

$$x_{cm} = \frac{m\frac{L}{2} + mL + m\frac{5L}{4}}{3m}$$

$$= \frac{11L}{12}$$

11) Kinetic energy $= \frac{1}{2}mv^2 = K = as^2$

or, $mv^2 = 2as^2$

Centripetal force $= \frac{mv^2}{R} = \frac{2as^2}{R}$

12)

Given, $a_c = k^2rt^2 \Rightarrow \frac{v^2}{r} = k^2rt^2$

$\Rightarrow V = krt$

and $a_T = \frac{dv}{dt} = kr$
Tangential force $F_T = ma_T = mkr$
Power imparted by $F_T = F_T V = (mkr)\,(krt)$
$= mk^2r^2t$
Power imported by $f_c = 0$
Net power $P = mk^2r^2t$

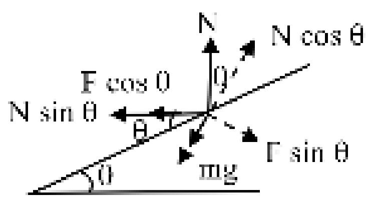
13)

$T_1 - T_2 = m\omega^2 ℓ$
$T_2 = m\omega^2 2ℓ$

14) $\omega_{AO} = \frac{(V_{AO})\perp}{r_{AO}} = \frac{V_0/\sqrt{2}}{\sqrt{4d^2 + 4d^2}} = \frac{V_0/\sqrt{2}}{2\sqrt{2}d} = \frac{V_0}{4d}$

15)

Centripetal force : $F_c = \frac{mv^2}{r}$



Balance vertical forces
$N\cos\theta = mg + \mu_s N\sin\theta$
$N(\cos\theta - \mu_s\sin\theta) = mg$
$N = \frac{mg}{\cos\theta - \mu_s \sin\theta}$
Balance horizontal forces to provide centripetal force:
$N\sin\theta + \mu_s\cos\theta = m^2_{max}$
$N(\sin\theta + \mu_s\cos\theta) = m^2_{max}$
$\frac{mg(\sin\theta + \mu_s\cos\theta)}{(\cos\theta - \mu_s\sin\theta)} = \frac{mv_{max}^2}{r}$
$V_{max} = \sqrt{\frac{gr(\sin\theta + \mu_s\cos\theta)}{\cos\theta - \mu_s\sin\theta}}$
$V_{max} = \sqrt{\frac{gR(\mu_s + \tan\theta)}{(1 - \mu_s\tan\theta)}}$

16) Given v = 20m/s
R = 100m
$a_c = \frac{v^2}{R} = \frac{(20)^2}{100} = 4\text{m/s}^2$
$a_t = \frac{dv}{dt} = 3\text{m/s}^2$

$a = \sqrt{a_c^2 + a_t^2} = \sqrt{4^2 + 3^2} = 5 m/s^2$

17)

The maximum speed (v_{max}) with which a vehicle can travel on a banked circular road is given by:

$v_{max} = \sqrt{\dfrac{rg\,(1 - \mu \tan\theta)}{\tan\theta + \mu}}$

$v_{max} \propto \sqrt{r}$

Percentage Increase = $\dfrac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100\%$

Percentage Increase = $\dfrac{v_2 - v_1}{v_1} \times 100\%$

$= \left(\dfrac{v_2}{v_1} - 1\right) \times 100\%$

$= \left(\sqrt{\dfrac{r_2}{r_1}} - 1\right) \times 100\%$

$= \left(\sqrt{\dfrac{4r}{r}} - 1\right) \times 100\%$

= 100%

18) $T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{\ell}{g}}$

$\Rightarrow \dfrac{2\pi}{2} = 2\pi\sqrt{\dfrac{\ell}{g}}$

$\dfrac{1}{4} = \dfrac{\ell}{g}$

$\ell = \dfrac{g}{4} = 2.5$ m

19) From equation : k = 5 .

As at t = 0.5 second, the particle reaches the origin for the first time.

Substituting : t = 0.5 in both equations

$x = 3 + 3\cos\left(\pi \cdot \dfrac{1}{2}\right) = 3 \rightarrow$ not satisfied

& $x = 3 + 3\cos\left(2\pi \cdot \dfrac{1}{2}\right) = 0 \rightarrow$ satisfied.

Hence , the equation of particle is :

20) Time period for half part :

$T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1}{g}} = \frac{2\pi}{\pi} = 2\,\text{sec}.$

So 2° part will be covered in a time t = $\frac{T}{2}$= 1s.
For the left 1° part : θ = θ_0 sin (wt)
⇒ 1° = 2° $\sin\left(\frac{2\pi}{T} \times t\right)$
$\Rightarrow \frac{1}{2} = \sin\left(\frac{2\pi}{2} \times t\right)$
⇒ $\frac{\pi}{6}$ = π × t ⇒ t = 1/6 sec.
Total time = $\frac{T}{2}$ + 2t ⇒ 1+2 × $\frac{1}{6}$=1+$\frac{1}{3}$ = $\frac{4}{3}$ sec.

21)

Theory

22) **1. Question Explanation (in 30 words):**
This problem involves a particle executing Simple Harmonic Motion (SHM). We are asked to find the displacement at which the kinetic energy (K.E.) and potential energy (P.E.) are equal.

2. Concept-Based (Name of Subtopic in 30 words):
This problem uses the concepts of Kinetic Energy (K.E.) and Potential Energy (P.E.) in Simple Harmonic Motion (SHM). The total energy is conserved, with K.E. and P.E. exchanging at various displacements.

3. Formula Used:
Kinetic Energy (K.E.): K.E. = $\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 = \frac{1}{2}mw^2(A^2 - x^2)$
Potential Energy (P.E.): P.E. = $\frac{1}{2}kx^2$
Where:
m is mass,
ω is angular frequency,
A is amplitude,
2 is displacement,
k is spring constant.

4. Calculation (in Brief):
In SHM, the total mechanical energy is constant and is given by:

$$E = \frac{1}{2}kA^2$$

At any displacement 2, the K.E. and P.E. are:

$$\text{K.E.} = \frac{1}{2}k(A^2 - x^2)$$

$$\text{P.E.} = \frac{1}{2}kx^2$$

For K.E. to equal P.E.,

$$\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$$

Simplifying:

$A^2 - x^2 = x^2$

$A^2 - 2x^2$

$x = \frac{A}{\sqrt{2}}$

Answer: The displacement at which K.E. equals P.E. is $x = \frac{A}{\sqrt{2}}$

23)

$k_1 = 3k$

$k_2 = \frac{3k}{2}$

$k_{eq} = \frac{9k}{2}$

$T = 2\pi\sqrt{\frac{2m}{9k}}$

24) Silence to next silence path difference is λ

40 – 10 = 30 cm = λ

n = 1200 Hz

V = nλ

V = 1200 × 30 cm/sec

25) $\frac{3V}{4\ell_1} = \frac{3V}{2\ell_2}$

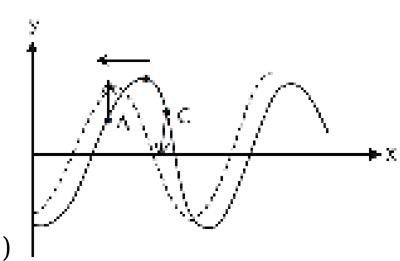
$\frac{\ell_1}{\ell_2} = \frac{2}{4} = 1 : 2$

26) ω = 2π rad/s, A = 10m, $K = \frac{\pi}{2}$

$v_{max} = A\omega = 10 \times 2\pi = 20\pi$ m/s

$v = \frac{\omega}{K} = \frac{2\pi}{\frac{\pi}{2}} = 4$ m/s

Ratio = $\frac{v_{max}}{v} = 5\pi$

27) 

28) ℓ_1 = 32.1 cm ℓ_2 = 99.2 cm

$e = \frac{\ell_2 - 3\ell_1}{2} = \frac{99.2 - 3(32.1)}{2} = \frac{2.9}{2} = 1.45\text{cm}$

$V = n\lambda \Rightarrow 332 = n[2(\ell_2 - \ell_1)]$

$\Rightarrow n = \frac{332}{2(99.2 - 32.1)} \times 10^{-2}$ = 247. 3 Hz

29) m = 4 gm, ℓ = 80cm

$\mu = \frac{m}{\ell}$

$= \frac{4 \times 10^{-3}}{80 \times 10^{-2}} = \frac{1}{200}$ kg/m

$V = \sqrt{\frac{T}{(\mu)}} = \sqrt{\frac{50}{1/200}}$ =100m/s

$f = \frac{3v}{2\ell} = \frac{3 \times 100 \times 100}{2 \times 80}$

= 187.5 Hz

30) $\because V_{0_C} > V_{0_B} > V_{0_A}$

$\therefore V_C > V_B > V_A$

$\lambda_C < \lambda_B < \lambda_A$

31) $P = \frac{nhc}{\lambda}$ Power of light = $\frac{15 \times 10^{-3} \times 10}{100}$

$n = \frac{P\lambda}{hc}$ $= 15 \times 10^{-4}$ w = 300

$= \frac{15 \times 10^{-4} \times 440 \times 10^{-9}}{2 \times 10^{-25}}$ per sec

$= 330 \times 10^{13} \times 60$ per min.

$= 1.98 \times 10^{17}$ per min.

32) $K_{max} = \left[\frac{12400}{5000} - 1.2\right]$ eV

$K_{max} = 1.28$ eV and $K_{min} = 0$

On reaching anode

$K'_{max} = (1.28 + 4) = 5.28$ eV

$K'_{min} = (0 + 4) = 4$ eV

33) $E_3 - E_1 = 12.09$ eV (in H atom)

$eV_0 = h\nu - \phi_0$

$\phi_0 = (12.09 - 8.09) \times 1.6 \times 10^{-19}$

$\nu^0 = \frac{4 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} \approx 10^{15}$ Hz

34)

Result of PEE experiment

35) $K_1 = \frac{hC}{\lambda} - \phi_0$

$K_2 = \frac{3hC}{\lambda} - \phi_0$

$K_2 = 3[K_1 + \phi_0] - \phi_0$

$K_2 = 3K_1 + 2\phi_0$

so $K_2 > 3K_1$

36)

Basic Theory

37)

In transverse electric field after time t speed v = $\sqrt{V_0^2 + \left(\frac{eC_0}{m}t\right)^2}$ so λ decreases increases.

In transverse electric field path of changed particle is parabola.

38) $V = \sqrt{2gh}$ = same for both

$\lambda = \frac{h}{mV}$

$\frac{\lambda_1}{\lambda_2} = \frac{m_2}{m_1} = \frac{4}{2} = \frac{2}{1}$

39) $\lambda = \frac{\lambda}{\sqrt{2mE}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{169}{100}} = \frac{13}{10}$

$\frac{\Delta\lambda}{\lambda}\% = \left[\frac{10}{13} - 1\right] \times 100\%$

$= -\frac{300}{13} = -23\%$

Wavelength decrease by 23%

40) $mv = \frac{h}{\lambda}$

$v = \frac{h}{\lambda m}$

$\frac{1}{\lambda} = R(1)^2 \left[\frac{1}{2^2} - \frac{1}{4^2}\right]$

$$\frac{1}{\lambda} = \frac{3}{16}R$$
$$v = \frac{3}{16}\frac{Rh}{m}$$

41)

basics of nuclear force.

42) Z – 2 × 2 + 2 × 1 – 4 × 1 ⇒ Z – 6
and A – 2 × 4 ⇒ A – 8
Number of neutrons = A – 8 – [Z – 6] = A – Z – 2
$$\frac{N}{Z} = \frac{A-Z-2}{A-6}$$

43) COLM $R = R_0A^{1/3}$ $R_1:R_2 = 3:1$
$$\frac{m_1}{m_2} = \frac{V_2}{V_1} \qquad \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}}$$
$$\frac{m_1}{m_2} = \frac{27}{1} \qquad \frac{R_1}{R_2} = \left(\frac{27}{1}\right)^{\frac{1}{3}}$$

44) CoLM.
$(A-4)v_N = 4v$
$$v_N = \frac{4v}{(210-4)} = \frac{4v}{206}$$

45)

density of all nuclis 2.35×10^{17} kg/m^3

CHEMISTRY

46)

α-sulphur is soluble in CS_2.

47)

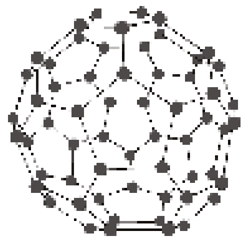
$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$ (Bond angle)

48) Formula based question

49) As sulphur is less electronegative than the oxygen thus H–S bond is weaker than OH and

H+ is easily released in case of H_2S as compare to H_2O.

50) Structure of C_{60}



It contain 20 hexagons 

and 12 pentagons 

so option 4 is incorrect.

51) P_4 + NaOH (aq.) $\rightarrow \overset{-3}{PH_3} + NaH_2\overset{+1}{PO_2}$
Oxidation state of P in PH_3
= –3 and 3 P–H bonds
Oxidation state of P in NaH_2PO_2
= +1 and 2 P–H bonds

52) I^- is reducing agent.

53)

Explanation -
The question presents an assertion and a reason related to the boiling points of noble gases.
Concept - Properties of Noble Gases.
Solution -
Assertion : Noble gases have very low boiling points due to weak interatomic forces.
Reason : Being monoatomic with complete shells, they only exhibit weak dispersion forces.
Both the assertion and the reason are true statements. The weak dispersion forces (explained in the reason) are the cause of the very low boiling points (stated in the assertion).
Final Answer - Option (1)

54)

Generated by Allie
Problem Statement: The question asks to identify which compound among the given options is a complex salt. It requires understanding the types of salts and recognizing the difference between complex salts and double salts.
Underlying Concept: Complex salts consist of complex ions where a central metal atom or ion is bonded to ligands, forming a coordination complex. Double salts are mixtures of two different salts that crystallize together but dissociate completely into their ions in aqueous solutions. The problem tests knowledge of coordination chemistry and the classification of salts.
Tips and Tricks: Remember that complex salts have complex ions that do not fully dissociate, while double salts separate into individual ions when dissolved.
Common Mistakes: Confusing double salts with complex salts since both may appear as

combined compounds. Not recognizing coordination complexes involving cyanide ligands.
Why Other Options Are Incorrect?: KClâ‹…MgClâ‚‚â‹…6Hâ‚‚O (carnallite), FeSOâ‚„â‹…(NHâ‚„)â‚‚SOâ‚„â‹…6Hâ‚‚O (Mohr's salt), and Kâ‚‚SOâ‚„â‹…Alâ‚‚(SOâ‚„)â‚ƒâ‹…24Hâ‚‚O (potash alum) are double salts. They dissociate completely in water into their constituent ions and do not form complex ions. Only 4KCNâ‹…Fe(CN)â‚‚ forms a complex ion, the ferrocyanide complex Kâ‚„[Fe(CN)â‚†], making it a true complex salt.

55)

Those ligands which have two or more than two denticity are called as chelating ligands. Ethylenediamine is bidentate ligand.

56)

			No. of unpaired e^-
$[CrF_6]^{4-}$	sp^3d^2	Cr^{+2}	4
$[MnF_6]^{4-}$	sp^3d^2	Mn^{+2}	5
$[Cr(CN)_6]^{4-}$	d^2sp^3	Cr^{+2}	2
$[Mn(CN)_6]^{4-}$	d^2sp^3	Mn^{+2}	1

57) $N(CH_3)_3$ and $P(CH_3)_3$ both are Lewis base and acts as ligand, However, $P(CH_3)_3$ has a p-acceptor character.

58)

A. EAN = 26 + 5 × 2 = 36
B. EAN = 28 + 4 × 2 = 36
C. EAN = 25 + 5 × 2 = 35
D. EAN = (25 + 1) + 5 × 2 = 36
In A, B, D EAN value comes out to be of nearest noble gas atomic number, which are stable. However in C which is not the case.

59) Metal - Carbon Bond order of x is more than that of y due to more entent of synergic bonding. Both the complexes have EAN = 36

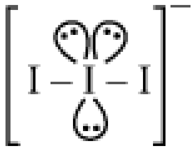
60) $[\overset{III}{Co}(en)_3]^{+3}\ SO_4^{-2}$ $[Co(en)_3]_2(SO_4)_3$

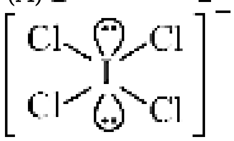
61) In PCl_3F_2 hybrid orbital's length at axial position is more than hybrid orbital's length at equitorial position.

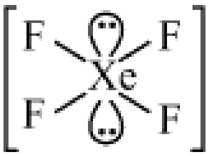
62) SO_2Cl_2, $SOCl_2$, P_4O_6, SF_4, $POCl_3$

63) σ^2_{1s}, σ^{x2}_{1s}, σ^2_{2s}, σ^{x2}_{2s}, $\sigma 2p_z^2$, $[\pi^2_{2px} = \pi^2_{2py}]$

$\left[\pi^{*1}{}_{2px} = \pi^{*1}{}_{2py}\right]$

64) (A) 

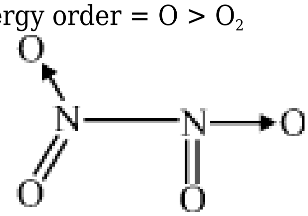
(B) 

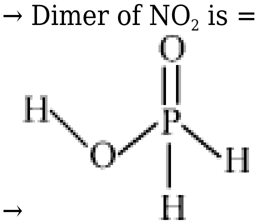
(C) 

65) Correct order
$BF_3 = BCl_3$ (bond angle)
$PbO > PbO_2$ (Oxidizing Property)

66) Due to large size of 'Xe' can show maximum polarisability which shows highest molecular interaction with H_2O. **OR** On moving down the group of 18, polarisability increases; He < Ne < Ar < Kr < Xe
Hence Xe with H_2O show maximum interaction.

67) → Ionisation energy order = $O > O_2$

→ Dimer of NO_2 is = 

→  is a monoprotic acid.

68) Refer notes.

69) (a) SF_4 – sp^3d hybridisation
(b) IF_5 – sp^3d^2 hybridisation
(c) NO_2^+ – sp hybridisation
(d) NH_4^+ – sp^3 hybridisation

70) O_2 (16 electrons)

$\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \sigma_{2p_z}^2$
$\pi_{2p_x}^2 = \pi_{2p_y}^2, \pi_{2p_x}^{*1} = \pi_{2p_y}^{*1}, \sigma_{2p_z}^*$
Bond order of $O_2 \Rightarrow 2$
Bond order of $O_2^- \Rightarrow 1.5$
Bond order of $O_2^{2-} \Rightarrow 1$
Bond order of $O_2^+ \Rightarrow 2.5$

71) SF_6 ⇒ 6 Identical bonds.
IF_7 ⇒ 5 Identical bonds.
PF_5 ⇒ 3 Identical bonds.
ClO_4^- ⇒ 4 Identical bonds.

72) (A) $\underset{\substack{\text{B.O.}=1.5\\ \text{Paramagnetic}}}{C_2^+} \rightarrow \underset{\substack{\text{B.O.}=2\\ \text{Diamagnetic}}}{C_2}$
(B) $\underset{\substack{\text{B.O.}=3\\ \text{Diamagnetic}}}{NO^+} \rightarrow \underset{\substack{\text{B.O.}=2.5\\ \text{Paramagnetic}}}{NO}$
(C) $\underset{\substack{\text{B.O.}=2\\ \text{Paramagnetic}}}{O_2} \rightarrow \underset{\substack{\text{B.O.}=2.5\\ \text{Paramangetic}}}{O_2^+}$
(D) $\underset{\substack{\text{B.O.}=3\\ \text{Diamagnetic}}}{N_2} \rightarrow \underset{\substack{\text{B.O.}=2.5\\ \text{Paramagnetic}}}{N_2^+}$

73) $[Co(NH_3)_5 Cl]Cl_2$ ⇒ give 3 mole ion

74)

V.F = 2 $FeCl_2 \rightarrow (Fe)_1$
1 Eq ⇒ $\frac{1}{2}$ mole
V.F = 3 $FeCl_3 \rightarrow (Fe)_2$
1Eq ⇒ $\frac{1}{3}$ mole
$n(fe)_1 : (Fe)_2$
$\frac{1}{2} : \frac{1}{3}$
3 : 2

75) **Calculation:**
1. Electrode Reactions and Valency:
$FeSO_4$: $Fe^{2+} + 2e^- \rightarrow Fe$ (valency = 2)
$Fe_2(SO_4)_3$: $Fe^{3+} + 3e^- \rightarrow Fe$ (valency = 3)
$Fe(NO_3)_3$: $Fe^{3+} + 3e^- \rightarrow Fe$ (valency = 3)
2. Equivalent Weights:
Equivalent weight of Fe in $FeSO_4$ = Atomic weight of Fe /2

Equivalent weight of Fe in $Fe_2(SO_4)_3$ = Atomic weight of Fe /3
Equivalent weight of Fe in $Fe(NO_3)_3$ = Atomic weight of Fe /3
3. Amount of Iron Deposited:
Since the same quantity of electricity is passed, the amount of iron deposited is proportional to the equivalent weight.
Let 'A' be the atomic weight of Fe.
Amount of Fe in $FeSO_4$ = k × (A/2) (where k is a constant)
Amount of Fe in $Fe_2(SO_4)_3$ = k × (A/3)
Amount of Fe in $Fe(NO_3)_3$ = k × (A/3)
Analysis of Statements:

A. (a) The amount of iron deposited in $FeSO_4$ and $Fe_2(SO_4)_3$ is equal.

This is incorrect. A/2 is not equal to A/3.

A. (b) The amount of iron deposited in $Fe(NO_3)_3$ is two third of the amount of Fe deposited in $FeSO_4$.

Amount of Fe in $Fe(NO_3)_3$ = k × (A/3)
Amount of Fe in $FeSO_4$ = k × (A/2)
(A/3) / (A/2) = 2/3. Therefore, this statement is correct.

A. (c) The amount of iron deposited in $Fe_2(SO_4)_3$ and $Fe(NO_3)_3$ is equal.

This is correct, as both have the same equivalent weight (A/3).
Final Answer:
Statement (b) and (c) are correct. Therefore, option 3 is correct.

76)
At anode : $Cd \longrightarrow Cd^{+2} + 2e$
At anode : $2H^{+} + 2e^{-} \longrightarrow H_2$

net cell reaction $\underset{(0.1\ M)}{Cd + 2H^{+}} \longrightarrow \underset{(0.1\ M)\ (0.5\ atm)}{Cd^{+2} + H_2}$

$$E_{cell} = 0.40 - \frac{0.06}{2}\log\frac{0.1 \times 0.5}{0.1 \times 0.1}$$
= 0.40 –(0.03 × log 5)
= 0.40 – 0.02
= 0.38

77) $E^0_{cell} = \left(E^0_{cathode} - E^0_{anode}\right)_{SRP}$
= 0.80 – (– 0.76) = 1.56

78)

At cathode : SRP of Cr^{+3} is higher
At anode : SOP of H_2O is higher

79)

For non-spontaneous process E^0_{cell} = –ve

80) $E^o_{cell} = E^o_{Fe^{+2}/Fe} - E^o_{sin^{+2}/sin}$
= – 0.44 – (– 0.14)
= – 0.30 volt

81)

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = (-0.403) - (-0.763) = +0.360\ V$

$Q = \frac{a_{Zn^{2+}}}{a_{Cd^{2+}}} = \frac{0.004}{0.2} = 0.02$

4. Apply the Nernst equation with $n = 2$: $E = +0.360 - \frac{0.0591}{2}\log(0.02)$ This matches the $E = +0.36 - \frac{0.059}{2}\log\left(\frac{0.004}{0.2}\right)$ second option gi

82)

1. The reaction quotient is $Q = \frac{[Zn^{2+}][Fe^{2+}]^2}{[Fe^{3+}]^2}$.
2. Increasing $[Fe^{2+}]$ increases the value of Q.
3. According to the Nernst equation, $E_{cell} = E^0_{cell} - (\text{ positive term }) \times \ln Q$.
4. As Q increases, lnQ increases, and the value subtracted from E^o_{cell} increase.
5. Therefore, E_{cell} decreases.

83)

$Pt_{(s)}|H_{2(g)}|H^+_{(eq.)}||Ag^+_{(eq.)}|Ag_{(s)}$

84) $E^o_{cell} = \frac{0.06}{2}\log K_{eq}$
$0.48 = 0.03 \log K_{eq}$
$\log K_{eq} = \frac{48}{3} = 16.0$
$K_{eq} = 10^{16}$

85)

NCERT XII Pg. # 89, Edition-2021-22
Density of H_2SO_4 decreases during discharge.

86)

On direction number of ions per unit volume decreases hence conductivity decreases.

87)

$E_{cell} = (E_{RP})_{cathode} - (E_{RP})_{Anode}$
$= (E_{RP})_{Cathode} + (E_{OP})_{Anode}$
$(\because E_{RP} = - E_{OP})$
$= (E_{OP})_{Anode} - (E_{OP})_{Cathode}$

88) $\lambda^{o}_{M_{NH_4OH}} = \lambda^{o}_{M_{NH_4^+}} + \lambda^{o}_{M_{OH^-}}$
$= \lambda^{o}_{M_{NH_4^+Cl^-}} + \lambda^{o}_{M_{N\overset{+}{a}OH^-}} - \lambda^{o}_{M_{Nacl}}$

89)

$Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$
$E_1 = E° - \frac{0.00591}{2} \log \frac{|Zn^{+2}|^1}{|Cu^{+2}|^1}$
$= E° - \frac{0.0591}{2} \log \frac{10^{-2}}{1} = E° + 0.0591$
$E_2 = E° - \frac{0.0591}{2} \log \frac{1}{10^{-2}} = E° - 0.0591$
$E_1 > E_2$

90)

Mobility of cation and anions of salt should be same or nearly same so option (3) wrong.

BIOLOGY

91)

NCERT-XI Pg#205

92) NCERT Pg. # 210, 211

93) NCERT Pg. # 208, 209, 210

94) NCERT pg. # 209, 212

95) NCERT Pg. # 209, 211, 212

96)

NCERT-XI Pg#205

97) NCERT Pg. # 291

98) NCERT Pg. # 213, 214

99)

NCERT-XI Pg#206,207,209,213

100) Correct Answer:

A. Option 2 - Ureotelic _________Insects (Incorrect Pair)

Explanation :

A. Uricotelic organisms excrete uric acid (e.g., Birds, Reptiles, Insects).
B. Ureotelic organisms excrete urea (e.g., Mammals like Elephants, Amphibians like Frogs).
C. Ammonotelic organisms excrete ammonia (e.g., Tadpoles, Bony Fishes, Amphibians).
D. Insects are uricotelic, not ureotelic, so Option 2 is incorrect.

101)

NCERT-XI Pg#208

102)

NCERT-XI Pg#212

103) NCERT Pg. # 209, 211, 212

104) NCERT Pg. # 206

105) NCERT Pg. # 206

106) NCERT Pg. # 212, 213

107) NCERT Pg. # 213

108)

NCERT-XI Pg#209

109)

NCERT-XI Pg#210,211

110) **Correct Answer: Option 3 - Urethral sphincter relaxes, urinary bladder contracts**
Explanation :

A. Micturition is the process of expelling urine from the bladder.
B. When the urinary bladder fills, stretch receptors send signals to the CNS.
C. The CNS triggers contraction of the bladder's smooth muscles and relaxation of the urethral sphincter, allowing urine to pass out.

Thus, during micturition, the urinary bladder contracts, and the urethral sphincter relaxes.

111)

NCERT-XI Pg#210, Module

112) NCERT Pg. # 206, 207

113) NCERT Pg. # 206,207

114) NCERT Pg. # 206, 207

115) NCERT Pg. # 206, 207

116)

NCERT Pg. # 207

117) NCERT Pg. No. # 207, 208

118)

NCERT-XI Pg#207

119)

NCERT-XI Pg#209

120)

NCERT-XI, Pg.# 242, 247

121)

NCERT-XI Pg#218

122)

NCERT-XI Pg#221

123) NCERT, Pg. # 304

124) **Question Explanation:** Evaluate statements about diseases.

Concept: Bone and Muscle disorders.

Solution: Statement I: Osteoporosis is a disorder characterized by **decreased** bone mass, not increased, which leads to an increased risk of fractures. Therefore, this statement is false. **Statement II: Myasthenia gravis** is an autoimmune disease where the body's immune system attacks the **motor-end plate** (neuromuscular junction), leading to muscle weakness and fatigue. This statement is true.

Final Answer: Statement-I is false but statement-II is true

125) NCERT Pg. 222

126) NCERT Pg. 225,226

127) NCERT, Pg. # 227

128) NCERT Pg. # 225-226

129)

NCERT-XI Pg#221,222

130) XI NCERT Page No. 307

131)

NCERT-XI Pg#226

132)

NCERT-XI Pg#224,225

133)

NCERT-XI Pg#219,220

134)

NCERT-XI Pg#219

135)

NCERT-XI Pg#224

136)

NCERT-XIth, Pg. 134

137) NCERT-XIth, Pg. # 135

138) NCERT-XIth, Pg. # 136

139) NCERT-XIth, Pg. # 137

140) NCERT-XIth, Pg. # 138

141)

NCERT-XIth, Pg. # 143, 146

142)

NCERT-XIth, Pg. # 147

143)

NCERT-XIth, Pg. # 150

144)

NCERT XI Pg. # 149, 150

145) NCERT-XIth, Pg. # 137

146)

NCERT-XIth, Pg. # 143, 146

147) NCERT-XI Pg. # 135

148) NCERT-XIth, Pg. # 145

149)

NCERT-XIth, Pg. # 141

150) NCERT-XIth, Pg. # 154

151) NCERT-XIth, Pg. # 154-155

152)

NCERT-XIth, Pg. # 156
Question Explanation : The question asks from which molecule in glycolysis are electrons (redox equivalents) removed to form NADH.
Concept: This question is based on Glycolysis
Solution:
PGAL (Phosphoglyceraldehyde), also known as G3P (Glyceraldehyde-3-phosphate), is oxidized. During this step, 2 electrons and 1 proton are transferred to NAD^+, forming NADH (the redox equivalent).
So, the correct answer is: 1. PGAL
Final Answer: Option (1)

153) NCERT-XIth, Pg. # 158

154)

NCERT-XIth, Pg. # 159
In the Krebs cycle, $FADH_2$ is produced during the conversion of Succinic Acid to Fumaric Acid, catalyzed by the enzyme Succinate Dehydrogenase. Fumaric acid is then converted to Malic Acid, but no $FADH_2$ is released in that step.

155)

NCERT-XIth, Pg. # 160

156)

NCERT XI, Pg. # 162, 163

157)

NCERT XI Pg.# 163, 164

158) NCERT-XIth, Pg. # 157

159) NCERT-XIth, Pg. # 157

160)

NCERT-XIth, Pg. # 161

161) NCERT-XIth, Pg. # 156,158,159,160

162) NCERT-XIth, Pg. # 158,159

163)

NCERT-XIth, Pg. # 160
Explain Question: It asks how many cytochromes from the list are part of the respiratory electron transport system (ETS).
Concept : This question is based on ETS (Respiration)
Solution:
Cytochromes involved in mitochondrial respiratory ETS:
Cytochrome b
Cytochrome c_1
Cytochrome c
Cytochrome a
Cytochrome a_3
So, the correct answer is: 2. Four
Final answer: Option (2).

164)

NCERT-XIth, Pg. # 175

165) NCERT-XIth, Pg. # 168

166)

NCERT-XIth, Pg. # 169

167)

NCERT-XIth, Pg. # 170

168) NCERT-XIth, Pg. # 174, 175

169)

NCERT-XIth, Pg. # 176

170)

NCERT-XIth, Pg. # 177
Ethylene
Femaleness (Feminising effect) in cucumbers. (Promotes female flowers to increase the production)

171)

NCERT-XIth, Pg. # 177, 178

172) NCERT-XIth, Pg. # 172, 173

173)

NCERT-XIth, Pg. # 175, 176, 177, 178

174) NCERT-XIth, Pg. # 167

175)

NCERT-XIth, Pg. # 167

176) NCERT-XIth, Pg. # 172, 173, 174

177)

NCERT XI, Pg. # 175, 176, 177
[2, 4-D, NAA & Kinetin]

178) NCERT XI Pg. # 145, 150, 151

179) NCERT XI Pg. # 158, 159

180) NCERT XI Pg. # 174